Fenil Miteshkumar Modi

Ahmedabad, Gujarat, India | +91 7874160204 | fenilmmodi@gmail.com | linkedin.com/in/fenilmodi823 | github.com/fenilmodi823

Profile

Computer Engineering graduate (degree conferral pending) seeking research positions in **space situational awareness** and **large-scale scientific data processing**. First author and presenter of an IEEE-published paper on probabilistic conjunction assessment; experienced across the full analysis chain from acquisition and preprocessing through probabilistic modelling, uncertainty quantification and validation. Available for international relocation.

Publication

Probabilistic Space Debris Conjunction Assessment using Machine Learning and Covariance Intersection Analysis — **Published**

- **First author and presenter**, with S. V. Khara, G. D. Tivari, J. Patel, P. Patel and G. Kumawat. *Proc.* ICSSIT 2026, IEEE — co-sponsored by the **IEEE Systems, Man and Cybernetics Society**. IEEE Xplore: 11656410 | DOI: [[DOI]]
- Showed that a nominal miss distance can remain statistically consistent with collision, as measured by overlap of 3σ covariance ellipses in the B-plane, by fusing state estimates of unknown correlation through covariance intersection and screening conjunctions with machine learning.

Experience

Hubbl — Technology Community Platform — India

Lead Backend & Data Engineer | Full-time — *Jul 2026 – Present*

- Defined the backend and data architecture at platform inception, as measured by 8+ core services and the PostgreSQL schema underpinning all community, event and speaker records, by setting service boundaries and data contracts.
- Turned four separate regional user-group datasets into one queryable relational schema, as measured by full consolidation into a single store, by designing the data model and ingestion path.
- Kept event and community data current without manual entry, as measured by mail-sync and scraping jobs running unattended on schedule, by building Node-based background workers; owned the registration pipeline for a 120-capacity national analytics event with the India AI + Tableau User Group and Queen's University Belfast.

ORCAS — Orbital Risk and Conjunction Assessment System — Ahmedabad, India

Team Lead & Lead Analyst | Silver Oak University — *Jan 2024 – Present*

- Cut catalogue preparation from a manual multi-hour task to a single automated pass, as measured by end-to-end runtime over a ∼10,000-object TLE catalogue, by building a Python ingestion pipeline that validates checksums and resolves epoch drift.
- Replaced point estimates with quantified confidence, as measured by 95% confidence intervals on probability of collision, by Monte Carlo sampling over positional covariance in a batch-parallel workload.
- Established propagator correctness before publication, as measured by position residuals against published reference ephemerides, by cross-validating SGP4 against independent reference outputs.
- Kept a six-member team reproducible, as measured by full reconstruction of every published figure from versioned source, by instituting Git-based code review and parameterised notebooks.

Selected Project

Quantum-Random Key Generation — Statistical Evaluation | *Python, Qiskit, SciPy* — 2023

- Determined whether quantum-derived keys outperform pseudo-random streams, as measured by pass rates across the 15 statistical tests of **NIST SP 800-22**, by designing a controlled comparison over matched-length key sets.
- Characterised diffusion behaviour with stated significance, as measured by χ^2 goodness-of-fit and avalanche-effect distributions, by implementing entropy and bit-correlation analysis in NumPy/SciPy.

Education

Silver Oak University — Ahmedabad, India

B.Tech., Computer Engineering — coursework completed, degree conferral pending — *Aug 2022 – Aug 2026*

- **Coursework:** Statistical Methods, Machine Learning, Data Structures & Algorithms, Cryptography, Computer Networks, Database Systems, Scientific Computing.

Technical Skills

Scientific Computing & Analysis: Python (NumPy, SciPy, Pandas), exploratory data analysis, time-series analysis, hypothesis testing, Monte Carlo methods, uncertainty quantification, numerical integration
Large-Scale Data Processing: batch pipelines, catalogue-scale datasets, scheduled background jobs, parallel workloads, performance profiling, Linux/CLI, HPC concepts
Orbital & Space Data: SGP4/SDP4, TLE handling, Skyfield, ephemeris validation, conjunction screening, covariance intersection, B-plane geometry, coordinate frames
Machine Learning: Scikit-learn, TensorFlow, PyTorch; cross-validation, residual analysis, model selection
Backend & Data Engineering: REST API design, schema design, query optimisation, JWT auth, Node.js, Express.js, PostgreSQL, Prisma, FastAPI, MongoDB, React
Research Practice: Git/GitHub, reproducible analysis, LaTeX, Jupyter, Matplotlib, Tableau, technical documentation
Languages: English (full professional), Hindi (native), Gujarati (native)